

Linear Algebra (MT1004)

Final Exam

Date: 1st January, 2024

Total Time (Hrs): 3

Course Instructor(s)

Total Marks: 90

Total Questions: 5

Dr. Akhlaq Ahmad
Dr. Tayyaba Naz
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Dr. Sonia Hanif
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Ms. Maria Shabir

Section

Student Signature

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Instruction/Notes:

1. Programmable calculators are not allowed.
2. Wrong calculation work found (if any) at a step will not be further marked. Marks will be awarded till the correct calculations.
3. Attempt all question parts together. Question attempted in separate parts will not be marked.

CLO #1: Use concept of elementary row operations to find the inverse of square matrices, determinant of a matrix and solving the system of linear equations.
CLO #5: Express a linear transformation graphically using matrices and to solve problems.

Application in Computer Graphics:

Question#1: [3+2 +3+5+2+5 marks]

- a) Use **Inversion Algorithm** to find the Inverse of matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}$. Write down A^{-1} as a product of elementary matrices $A^{-1} = E_k E_{k-1} \dots E_3 E_2 E_1$.
- b) Verify that $A = E_1^{-1} E_2^{-1} E_3^{-1} \dots E_{k-1}^{-1} E_k^{-1}$ for some k .
- c) Sketch the image of the triangle with vertices $(0, 0)$, $(1.5, 2)$, $(3, 1)$ under multiplication by the invertible matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}$.
- d) Discuss the **Geometric Effect** on the triangle of multiplication by the given matrix A , using the following steps:
- i) Show the effect of $E_1^{-1} E_2^{-1} E_3^{-1} \dots E_{k-1}^{-1} E_k^{-1}$ on the triangle with vertices $(0, 0)$, $(1.5, 2)$, $(3, 1)$ step by step.
 - ii) Show mathematically action of each elementary matrix on the end points of the edges and graphically show the output images at each step.
- e) Show that succession of shears, compressions, expansions, and reflections that obtained in part (d) produces the same image as obtained in part (c).
- f) Find an equation for the image of the line $y = -4x + 3$ under multiplication by the matrix

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}.$$

CLO #2: Properties of vectors in 2-space, 3-space and n-space and recognize vector spaces and/or subspaces to compute their bases and its dimension.

Question#2: [5 +5+10 marks]

- a) Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $u = (u_1, u_2)$ and $v = (v_1, v_2)$:

$$u + v = (u_1 + v_1 - 3, u_2 + v_2 - 3), \quad ku = (ku_1, 0).$$

- i) Show that Axiom 4 holds by producing a zero vector such that $u + (0) = u$ for $u = (u_1, u_2)$.
 ii) Show that Axiom 5 holds by producing a negative vector $(-u)$ such that $u + (-u) = 0$ for $u = (u_1, u_2)$.
 iii) Show that Axiom 8, i.e. $[(k + m)u = ku + mu]$ and Axiom 10 $[1 \cdot u = u]$, fail and hence that V is not a vector space under the given operations.

- b) Consider the bases $B = \{u_1, u_2\}$ and $B' = \{u_1', u_2'\}$ for R^2 , where $u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$u_1' = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \text{ \& } u_2' = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Find the transition matrix $P_{B \rightarrow B'}$.

$$\begin{bmatrix} 1/5 & 2/5 \\ 1/5 & -3/5 \end{bmatrix}$$

- c) For the given matrix

$$A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -2 & 2 \\ 3 & 6 & -3 & 3 \\ 4 & 8 & -4 & 4 \end{bmatrix}$$

- i) Find the basis for the row space of A .
 ii) Find the basis for the column space of A .
 iii) Find the basis for the null space of A and explain geometrically the solution space/sub-space/null space spanned by the basis for the null space of A . **Hint:** Write the solution space say " X " in Matrix column notation and then check cardinality of X to determine basis for null space.
 iv) Find rank and nullity for the given matrix A .

CLO #3: Perform Eigen Value analysis and use it to Diagonalize a matrix and/or find its powers.

Question#3: [5+2+8+5 marks]

- a) Find the geometric and algebraic multiplicity of each eigenvalue of the matrix A ,
 b) Determine whether A is diagonalizable. If A is diagonalizable, then find a matrix P that diagonalizes A .
 c) Prove that $P^{-1}AP = D$.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{bmatrix}$$

- d) Check that matrix A and $P^{-1}AP$ have same trace by using the definition of similarity invariants.

CLO #4: Identify inner product spaces and/or perform Gram Schmidt process/QR decomposition using inner products.

Question#4: [5+5 marks]

Suppose $A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & -1/2 \end{bmatrix}$

define the column vectors u_1, u_2 and u_3 as
 $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \& u_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

- a) Use the **Gram – Schmidt** process to find an orthogonal set of vectors $\{v_1, v_2, v_3\}$ and then find orthonormal set of vectors $\{q_1, q_2, q_3\}$. Discuss the geometry of Eigen Spaces corresponding to each Eigen value.

- b) Find the **QR-** decomposition of the given matrix. Also verify that $A = QR$ where,
 $Q = [q_1 \mid q_2 \mid q_3]$ consists of the column vectors obtained in part (a) and R is given below

$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \langle u_3, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \langle u_3, q_2 \rangle \\ 0 & 0 & \langle u_3, q_3 \rangle \end{bmatrix}$

Note: Consider standard dot product as standard inner product between vectors.

CLO #5: Express a linear transformation graphically using matrices and to solve problems.

General Linear Transformations:

Question # 5: [5+10+5 marks]

- a) Consider basis $S = \{v_1, v_2\}$ for R^2 , where $v_1 = (-2, 1)$ and $v_2 = (1, 3)$, and let $T: R^2 \rightarrow R^3$ be the linear transformation such that $T(v_1) = (-1, 2, 0)$ and $T(v_2) = (0, -3, 5)$. Find a formula for $T(x_1, x_2)$, and use that formula to find $T(2, -3)$.

- b) Let $T: R^2 \rightarrow R^3$ be the linear transformation defined by

$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -4x_1 \\ -x_1 + 2x_2 \\ -2x_1 + 5x_2 \end{bmatrix}$

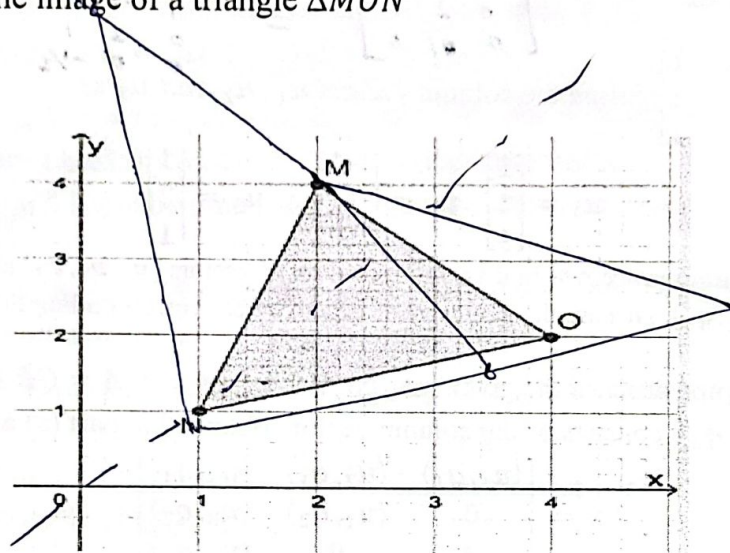
- i) Find the matrix for the transformation T i.e. $[T]_{B', B} = [[T(u_1)]_{B'} \mid [T(u_2)]_{B'}]$ relative to the basis $B = \{u_1, u_2\}$ and $B' = \{v_1, v_2, v_3\}$, where

$u_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, u_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, v_1 = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

Denote $T = [T]_{B', B}$ and find the following:

- ii) Find the Kernel of T i.e. $\text{Ker}(T)$.
iii) Find the Range of T i.e. $R(T)$.

c) The following is the image of a triangle $\triangle MON$



Find standard matrices for the mentioned below parts and plot the final transformed image of a given triangle.

- Expand by a factor of 2 in the x -direction
- Reflect the given triangle about line $y = x$

Good Luck!

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	Course:	Linear Algebra	Course Code:	MT-1004
	Program:	BS (CS, DS, SE)	Semester:	Fall 2022
	Duration:	3 Hours	Total Marks:	100
	Date:	31-12-2022	Weight	50%
	Section:	All	Page(s):	3
	Exam:	Final	Roll No:	
Name: _____				

Instruction/Notes:

Use of programmable calculators is not allowed. Exchange of stationary is strictly prohibited. Show complete working in all questions. Attempt all question parts together. Question attempted in separate parts will not be marked.

Q#8 is a BONUS Question. (You will not lose marks if unable to attempt).

Question#1: [5+5+5+5 marks, CLO#1, 5]

- Use Inversion Algorithm to find the Inverse of matrix $A = \begin{bmatrix} 0 & 2 \\ 4 & 0 \end{bmatrix}$.
- Write down A^{-1} as a product of elementary matrices $A^{-1} = E_k E_{k-1} \dots E_3 E_2 E_1$.
- Verify that $A = E_1^{-1} E_2^{-1} E_3^{-1} \dots E_{k-1}^{-1} E_k^{-1}$ for some k .
- Use the decomposition done in part (c) to discuss the Geometric Effect on the Unit Square of the elementary matrices.

Do the following steps for part (d):

- Show the effect of $E_1^{-1} E_2^{-1} E_3^{-1} \dots E_{k-1}^{-1} E_k^{-1}$ on the unit square step by step.
- Show the action of each elementary matrix mathematically and graphically.

Question#2: [5+5 marks, CLO #2]

- Find the area of the triangle in 3-space that has the given vertices.
 $P = (1, -1, 2), Q = (0, 3, 4), R = (6, 1, 8)$.
- Find the distance between the point $(2, 1, -3)$ and the plane $2x - y - 2z = 6$.

Question#3: [10 marks, CLO #3]

Find the no. of vectors and degree geometric and algebraic multiplicity of each eigenvalue of the matrix A ,

$$A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$$

and determine whether A is diagonalizable. If A is diagonalizable, then find a matrix P that diagonalizes A and find $P^{-1}AP$.

Question#4: [10 marks, CLO #03]

Suppose you are designing a simple video game where the player controls a shooter to hit a moving targets. If the shooter is an arrowhead whose vertices at any point is in the span of row vectors of A where

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 & 3 \\ -2 & 5 & -7 & 0 & -6 \\ -1 & 3 & -2 & 1 & -3 \\ -3 & 8 & -9 & 1 & -9 \end{bmatrix}$$

(Hint: Take Transpose of given matrix A)

- Find the basis for the row space of A consisting entirely of the row vectors from A .
- Use Dimension theorem to find the rank and nullity of A and A^t .

Question#5: [20 marks, CLO #04]

Suppose $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ define the column vectors u_1, u_2 and u_3 as

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \text{ \& } u_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

- Use Gram – Schmidt process to find the orthogonal set of vectors $\{v_1, v_2, v_3\}$ and then find orthonormal set of vectors $\{q_1, q_2, q_3\}$ by considering standard inner product between vectors.
- Considering standard inner product between the vectors find QR- decomposition of the given matrix. Also verify that $A = QR$ where,

$Q = [q_1 \mid q_2 \mid q_3]$ and R is given below

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \langle u_3, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \langle u_3, q_2 \rangle \\ 0 & 0 & \langle u_3, q_3 \rangle \end{bmatrix}.$$

Question#6: [5+5 marks, CLO #04]

- Find $\|u\|$ and $d(u, v)$ relative to the weighted inner product defined as $\langle u, v \rangle = 2u_1v_1 + 3u_2v_2$ on R^2 where, $u = (-3, 2)$ and $v = (1, 7)$.
- Considering the weighted inner product defined in part (a) check that whether the vectors are orthogonal.

Question#7: [10 marks, CLO #05]

Let $T: R^2 \rightarrow R^3$ be the linear transformation defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -x_1 \\ -3x_1 + 2x_2 \\ -4x_1 + 3x_2 \end{bmatrix}$. Find the matrix for the transformation T i.e. $[T]_{B', B} = [[T(u_1)]_{B'} \mid [T(u_2)]_{B'}]$ relative to the basis $B = \{u_1, u_2\}$ and $B' = \{v_1, v_2, v_3\}$, where

$$u_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, u_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Question#8: [5+5 marks, CLO-3, 5] (Bonus Question)

If $C = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, then

Do the following steps:

- a. Find a matrix P (consisting of the Eigen vectors of the matrix C) using Eigenvalues of C and show that $P^{-1}CP = D$. Also, find the dimension of Eigen Spaces associated with each Eigen value.
- b. Show that C and D represents same linear operator $T : R^2 \rightarrow R^2$ by showing $P^{-1}CP = D$, where $P = P_{B' \rightarrow B} = [[u'_1]_B \quad [u'_2]_B]$ and $P^{-1} = P_{B \rightarrow B'}$, $B' = \{u'_1, u'_2\}$, $B = \{e_1, e_2\}$, $u'_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ & $u'_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Here P & P^{-1} shows the transition matrices.

Linear Algebra (MT1004)

Date: May 27th 2024

Course Instructors

Dr Saeeda Zia

Ms Sarah Asghar

Final-Term Exam

Total Time: 3 Hours

Total Marks: 85

Total Questions: 8

22i-2505 SE-4A

Roll No

Section

JKS

Student Signature

Attempt all the questions. Attach question paper with answer sheet.

CLO # 1: Convert real life problems into system of linear equations and perform elementary row operations to solve them. (10 marks)

Q. No 1: Zainab, Ayesha, and Maya go to a jewelry store. Zainab buys 3 necklaces, 1 ring, and 2 bracelets and spends \$17. Ayesha buys 1 necklace, 4 rings, and 5 bracelets and spends \$31. Maya buys 6 necklaces, 2 rings, and 1 bracelet and spends \$19. Determine the system of linear equations from the given information where the unknown variables x , y and z represents necklaces, ring and bracelets respectively. How much does each type of jewelry cost? Use the Gauss Elimination method.

CLO #4: Analyse vectors and their properties in 2-space, 3-space and n-space. (10 marks)

Q. No 2: Determine whether the given planes are parallel. Also find the distance between them.
 $2x - y + z = 1$ and $2x - y + z = -1$

CLO #5: Recognize vector spaces and/or subspaces and compute their bases and its dimension. (5+5 marks)

Q. No 3: Given the set of vectors $v_1 = (1, 2, 3)$, $v_2 = (-4, 5, 6)$, $v_3 = (7, -8, 9)$.

a. Show that $\{v_1, v_2, v_3\}$ forms a basis for R^3 .

b. Find the coordinate vector of $w = (5, -12, 3)$ relative to the basis $S = \{v_1, v_2, v_3\}$.

CLO #6: Eigen Value analysis and use it to Diagonalize a matrix and find its powers. (10 marks)

Q. No 4: Find a matrix P that diagonalizes A , and check your work by computing $P^{-1}AP$.

$$\begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix}$$

and compute A^{10} .

CLO #7: Identify inner product spaces. (10 marks)

Q. No 5: Let $p = x + x^3$ and $q = 1 + x^2$. Find $\|p\|$ and $d(p, q)$ relative to the evaluation inner product P_3 at the stated sample points. $x_0 = -2, x_1 = -1, x_2 = 0, x_3 = 1$.

CLO #7: Identify inner product spaces and/or perform Gram Schmidt process. (10 marks)

Q. No 6: Let R^3 have the Euclidean inner product and use the Gram-Schmidt process to find the basis $\{u_1, u_2, u_3\}$ into an orthonormal basis where $u_1 = (1, 0, 0)$, $u_2 = (3, 7, -2)$, $u_3 = (0, 4, 1)$.

CLO #8: Express a linear transformation using matrices and to solve problems. (10 marks)

Q. No 7: Consider the basis $S = \{v_1, v_2, v_3\}$ for R^3 , $v_1 = (1, 2, 1)$, $v_2 = (2, 9, 0)$, and $v_3 = (3, 3, 4)$, and let $T: R^3 \rightarrow R^2$ be the linear transformation for which

$$T(v_1) = (1, 0), T(v_2) = (-1, 1), T(v_3) = (0, 1).$$

Find a formula for $T(x_1, x_2, x_3)$, and use that formula to find $T = (7, 13, 7)$.

CLO #8: Express a linear transformation using matrices and to solve problems. (5+5+5 marks)

Q. No 8: Given the standard matrix $A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}$ of a matrix transformation $T_A : R^3 \rightarrow R^3$.

- Determine T_A is an isomorphism.
- Find the range and Kernel of the transformation.
- Also find the dimension of row space and null space of A .

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National University of Computer and Emerging Sciences, Lahore Campus				
	Course:	Linear Algebra	Course Code:	MT104
	Program:	BS (CS, SE, DS)	Semester:	Fall
	Duration:	3 hours	Total Marks:	100
	Paper Date:	20-1-22	Weight	50-54%
	Section:	All	Page(s):	03
	Exam:	Final term	Roll No:	
Instruction/Notes:		Programmable calculators are not allowed. Show complete working in all questions. 54% wtg. is applicable to only those sections who had no quiz-3 due University Closure. Q#9 is BONUS question.		

Application in Computer Graphics

Question#1[05+05][CLO-1]: Use Elementary Matrices to find the Inverse of $A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$. Also verify that $A = E_1^{-1} E_2^{-1} E_3^{-1} \dots E_{k-1}^{-1} E_k^{-1}$ for some k .

Any other method used to evaluate the inverse will not be considered for marking.

Application in Computer Graphics

Question#2[2+5+5+5+3][CLO-1,5]: Discuss the Geometric Effect on the Unit Square of multiplication by the matrix $A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ using the following steps:

1. Decompose $A = E_1^{-1} E_2^{-1} E_3^{-1} \dots E_{k-1}^{-1} E_k^{-1}$ for some k .
2. Show the effect of $E_1^{-1} E_2^{-1} E_3^{-1} \dots E_{k-1}^{-1} E_k^{-1}$ on the unit square. Also show the action of elementary matrix (each) via diagram separately.
3. Show mathematically action of each elementary matrix on the end points of the edges.
4. Illustrate the geometric effects at each step.

Lines & Planes in R^3

Question#3(a) [05][CLO-2]: Find the vector and parametric equation of the plane in R^3 that passes through the origin and is orthogonal to $= (3, 1, -6)$.

3(b)[5] [CLO-2]: Find a vector parallel to the line of intersection of two planes $3x - 6y - 2z = 15$ & $2x + y - 2z = 5$.

Gram-Schmidt Process for Orthonormal basis

Question#4[20][CLO-4]: Suppose $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$ define the column vectors u_1, u_2 and u_3 as

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \text{ \& } u_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

$$\begin{aligned} 2 - \frac{1}{2} &= \frac{3}{2} = 3/2 \\ \frac{1}{2} - \frac{1}{5} &= 4/10 \\ -\frac{1}{2} - \frac{2}{5} &= -\frac{5+4}{10} = -\frac{9}{10} \end{aligned}$$

$$\begin{aligned} 2 - \frac{3}{2} &= \frac{1}{2} \\ \frac{4-3}{2} &= \frac{1}{2} \end{aligned}$$

4(a)[10]: Use the Gram – Schmidt process to find the orthogonal set of vectors $\{v_1, v_2, v_3\}$ and then an orthonormal set of vectors $\{q_1, q_2, q_3\}$ by considering standard inner product between the vectors.

4(b)[10]. Also find a matrix R and verify $A = QR$ where, $Q = [q_1 \mid q_2 \mid q_3]$ and R is given below

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \langle u_3, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \langle u_3, q_2 \rangle \\ 0 & 0 & \langle u_3, q_3 \rangle \end{bmatrix}$$

General Linear Transformations

Question#5 [2+2+2+2+2][CLO-5]: Let $T: R^2 \rightarrow R^3$ be the linear transformation defined by the formula

$$T(x_1, x_2) = (x_1 + 3x_2, x_1 - x_2, x_1)$$

- Find the Standard Basis Matrix (A) for the above transformation.
- Find the rank of A i.e. rank(A).
- Find the nullity of A i.e. null(A).
- Find the rank of the A^t i.e. rank(A^t).
- Find the nullity of the A^t i.e. null(A^t).

Question#6 [10][CLO-5]: Let $T: R^2 \rightarrow R^3$ be defined as $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} x_1 + 2x_2 \\ -x_1 \\ 0 \end{pmatrix}$. Find the matrix $[T]_{B', B} = [[T(u_1)]_{B'} \mid [T(u_2)]_{B'}]$ relative to the basis $B = \{u_1, u_2\}$ and $B' = \{v_1, v_2, v_3\}$, where

$$u_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, u_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}, v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}.$$

Question # 7 [2+2+2+2+2][CLO-5]: Let $T: R^3 \rightarrow R^3$ be a linear operator defined as

$T(x_1, x_2, x_3) = (0x_1 + x_2 - x_3, x_1 + 0x_2 + 2x_3, -1x_1 + x_2 + 0x_3)$ defined by $T(X) = AX$ as, where

$$A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}$$

- Check whether T is One to One.
- Check whether T is Onto.
- Find Kernel of T and Basis for Kernel of T .
- Find Range of T and Basis for Range of T .
- Find Null space of T and Row space of T .

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$q_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, q_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, q_3 = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$$

$$36 - \frac{16}{25}$$

Equivalence Theorem

Question#8[10]: STATE ONLY the Equivalent Statements (as much as you remember) for the $n \times n$ Matrix, if it's given that:

- a) A is invertible.
- b)

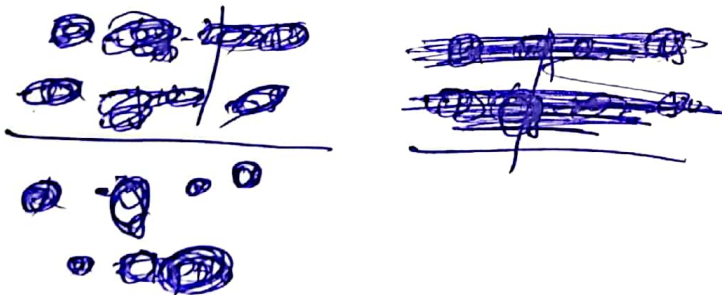
Note: For each equivalent statement one point will be given. Maximum points are 10.

Similarity of Operators (Bonus)

Question#9[05+05][CLO-3,5]: If $C = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, then

- a. Find a matrix P (consisting of the Eigen vectors of the matrix C) using Eigenvalues of C & show that $P^{-1}CP = D$. Also, find the dimension of Eigen Spaces associated with each Eigen value.
- b. Show that C and D represents the same linear operator $T : R^2 \rightarrow R^2$ by showing $P^{-1}CP = D$, where $P = P_{B' \rightarrow B} = [[u'_1]_B \quad [u'_2]_B]$ and $P^{-1} = P_{B \rightarrow B'}$, $B' = \{u'_1, u'_2\}$, $B = \{e_1, e_2\}$, $u'_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ & $u'_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Here P & P^{-1} represents the transition matrices.

Good Luck : Thank you 😊 (smile in pain)



Question # 1.

[3]

(a)

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}$$

To find the inverse using inversion algorithm, we have

$$\left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 2 & 9 & 0 & 1 \end{array} \right]$$

$$R_2 - 2R_1$$

$$\left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right]$$

$$E_1 = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$$

$$R_1 - 4R_2$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 9 & -4 \\ 0 & 1 & -2 & 1 \end{array} \right]$$

$$E_2 = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 9 & -4 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = E_2 E_1$$

$$= \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+8 & 0-4 \\ 0-2 & 0+1 \end{bmatrix} = \begin{bmatrix} 9 & -4 \\ -2 & 1 \end{bmatrix}$$

[2]

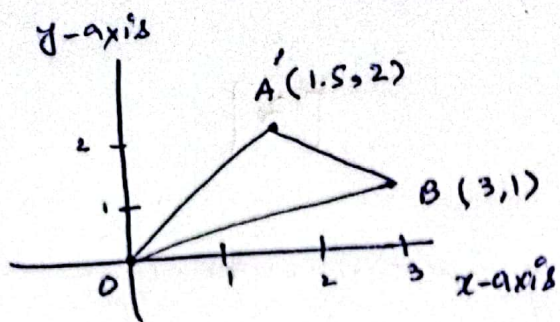
(b)

$$E_1^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$E_2^{-1} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

$$A = E_1^{-1} E_2^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 4 \\ 2+0 & 8+1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}$$

(c)



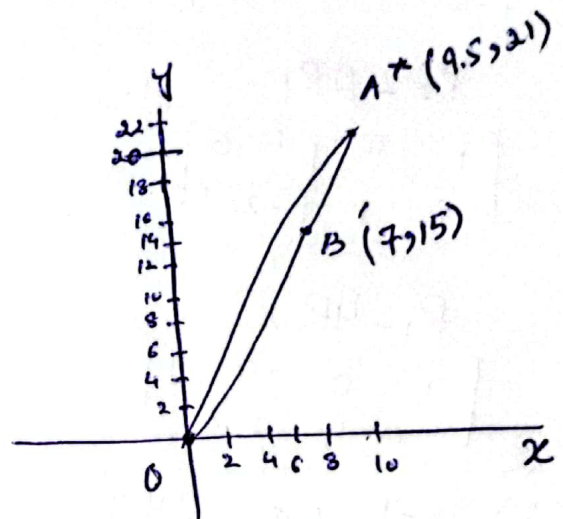
[3]

To find the image of triangle under multiplication,

$$\begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} = \begin{bmatrix} 9.5 \\ 21 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 15 \end{bmatrix}$$



[5]

(d)

$$(i) E_2^{-1} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

Shear in the x-direction with a factor of 4

$$E_1^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

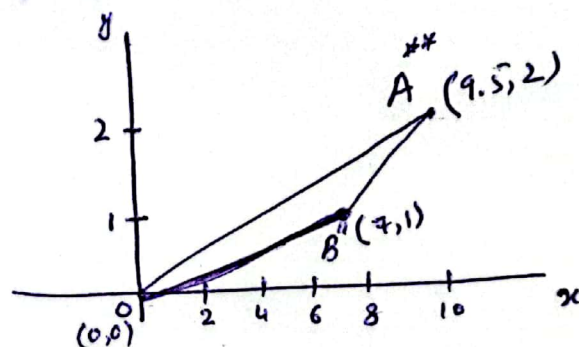
Shear in the y-direction by factor 2.

$$(u) \quad E_2^{-1} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

$$E_2^{-1} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$E_2^{-1} A' = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} = \begin{bmatrix} 1.5+8 \\ 0+2 \end{bmatrix} = \begin{bmatrix} 9.5 \\ 2 \end{bmatrix}$$

$$E_2^{-1} B = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3+4 \\ 0+1 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

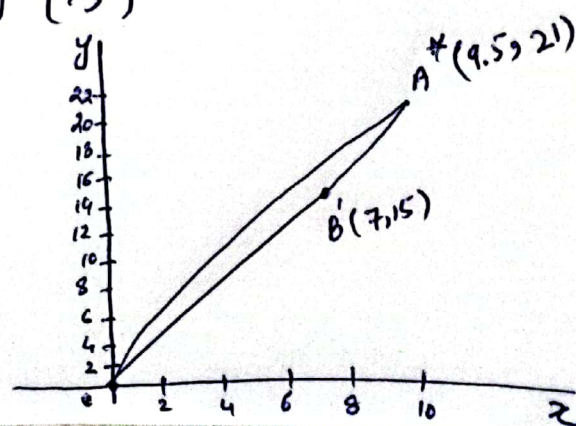


$$E_1^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$E_1^{-1} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$E_1^{-1} A^{**} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 9.5 \\ 2 \end{bmatrix} = \begin{bmatrix} 9.5+0 \\ 19+2 \end{bmatrix} = \begin{bmatrix} 9.5 \\ 21 \end{bmatrix}$$

$$E_1^{-1} B'' = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 1 \end{bmatrix} = \begin{bmatrix} 7+0 \\ 14+1 \end{bmatrix} = \begin{bmatrix} 7 \\ 15 \end{bmatrix}$$



[2]

(e)

We have seen that in part (b) the triangle is undergoing shear in x -direction and y -direction, i.e.,

$$= E_1^{-1} E_2^{-1}$$

$$= A \quad (\text{from part b})$$

It can also be observed clearly that when matrix A is applied on the vertices of triangle, we get the image obtained in part (c).

(f)

$$y = -4x + 3$$

[5]

Let (x, y) be the point on $y = -4x + 3$ and (x', y') be its image under multiplication by A

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -4 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 9x' - 4y' \\ -2x' + y' \end{bmatrix}$$

$$\Rightarrow x = 9x' - 4y', \quad y = -2x' + y'$$

Substitute these expressions in $y = -4x + 3$

$$\begin{aligned} -2x' + y' &= -4(9x' - 4y') + 3 \Rightarrow -2x' + y' = -36x' + 16y' + 3 \\ 16y' - y' &= 36x' - 2x' - 3 \Rightarrow 15y' = 34x' - 3 \end{aligned}$$

$$\boxed{y' = \frac{34}{15}x' - \frac{1}{5}} \rightarrow \text{Image of the given line.}$$

Q No. 2 (a)

$$u+v = (u_1+v_1-3, u_2+v_2-3)$$

$$ku = (ku_1, 0)$$

Part ii)

$$u + 0 = u$$

$$\text{Let } 0 = (a_1, a_2)$$

$$u + 0 = (u_1 + a_1 - 3, u_2 + a_2 - 3) = (u_1, u_2)$$

$$\Rightarrow u_1 + a_1 - 3 = u_1 \quad | \quad u_2 + a_2 - 3 = u_2$$

$$\Rightarrow a_1 - 3 = 0$$

$$a_2 - 3 = 0$$

$$a_1 = 3$$

$$a_2 = 3$$

$$\Rightarrow 0 = (a_1, a_2) = (3, 3)$$

$\Rightarrow$ Identity element exist.

axiom 4 holds.

Q 2 (a)

Part (ii) For $u = (u_1, u_2)$

$$u + (-u) = 0$$

Here $0 = (3, 3)$

~~Let~~ Let $-u = (b_1, b_2)$ be the additive inverse of u , Then

$$u + (-u) = (u_1 + b_1 - 3, u_2 + b_2 - 3) = (3, 3)$$

$$\Rightarrow \begin{array}{l|l} u_1 + b_1 - 3 = 3 & u_2 + b_2 - 3 = 3 \\ b_1 = 6 - u_1 & b_2 = 6 - u_2 \end{array}$$

$$\Rightarrow -u = (6 - u_1, 6 - u_2)$$

Hence additive inverse $(-u)$ exist.

Q. No. 2(a)

Part (iii)

Axiom 8

$$(K+m)u = Ku + mu.$$

L.H.S $(K+m)u = (K+m)(u_1, u_2)$

$$= ((K+m)u_1, 0)$$

$$= (Ku_1 + mu_1, 0) \text{ --- (1)}$$

R.H.S $Ku + mu = K(u_1, u_2) + m(u_1, u_2)$

$$= (Ku_1, 0) + (mu_1, 0)$$

$$= (Ku_1 + mu_1 - 3, 0 + 0 - 3)$$

$$= (Ku_1 + mu_1 - 3, -3) \text{ --- (2)}$$

From (1) and (2)

$$(K+m)u \neq Ku + mu$$

Axiom 8 fails.

Axiom 10

$$1 \cdot u = u.$$

$$\text{L.H.S.} = 1 \cdot u = 1 \cdot (u_1, u_2)$$

$$= (1 \cdot u_1, 0)$$

$$= (u_1, 0) \neq (u_1, u_2)$$

$$\neq u.$$

Hence, axiom ~~10~~ 10 fails.

Q 2 (b)

$$B = \{u_1, u_2\}$$

$$u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$B' = \{u'_1, u'_2\}$$

$$u'_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, u'_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Find (i) $P_{B \rightarrow B'}$

(ii) $P_{B' \rightarrow B}$

Sol: For part (i)

$$[B' | B] = [\text{New basis} | \text{old basis}]$$

$$= \left[\begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array} \right]$$

R_{12}

$$= \left[\begin{array}{cc|cc} 1 & -1 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{array} \right]$$

$R_2 - 3R_1$

$$= \left[\begin{array}{cc|cc} 1 & -1 & 0 & 1 \\ 0 & 5 & 1 & -3 \end{array} \right]$$

$\frac{1}{5} R_2$

$$= \left[\begin{array}{cc|cc} 1 & -1 & 0 & 1 \\ 0 & 1 & 1/5 & -3/5 \end{array} \right]$$

$R_1 + R_2$

Hence

$$P_{B \rightarrow B'} = \begin{bmatrix} 1/5 & 2/5 \\ 1/5 & -3/5 \end{bmatrix} = \left[\begin{array}{cc|cc} 1 & 0 & 1/5 & 2/5 \\ 0 & 1 & 1/5 & -3/5 \end{array} \right]$$

Q 2c

$$A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -2 & 2 \\ 3 & 6 & -3 & 3 \\ 4 & 8 & -4 & 4 \end{bmatrix}$$

$R_2 - 2R_1$

$R_3 - 3R_1$

$R_4 - 4R_1$

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

①

Now

Basis for row space = $(1, 2, -1, 1)$

Basis for the column space = $(1, 2, 3, 4)$

For Null space

$$x + 2y - z + w = 0$$

$$x = -2y + z - w$$

$$\text{Let } y = t_1$$

$$z = t_2$$

$$w = t_3$$

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = t_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + t_3 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Basis for the ~~sub~~ null space

$$\{(-2, 1, 0, 0), (1, 0, 1, 0), (-1, 0, 0, 1)\}$$

From (1)

$$\text{Rank} = 1$$

$$\text{nullity} = 3$$

Q#3(a) Find Eigenvalues

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{bmatrix}$$

Sol:-

Here $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Multiplying by λ

$$\lambda I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

We know that

$$\det(\lambda I - A) = 0$$

$$\begin{aligned} \text{So, } \lambda I - A &= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{bmatrix} \\ &= \begin{bmatrix} \lambda-2 & 0 & 0 \\ -1 & \lambda-4 & 1 \\ 2 & 4 & \lambda-4 \end{bmatrix} \end{aligned}$$

$$\det(\lambda I - A) = \begin{vmatrix} \lambda-2 & 0 & 0 \\ -1 & \lambda-4 & 1 \\ 2 & 4 & \lambda-4 \end{vmatrix} = 0$$

Expanding by 1st Row

$$\Rightarrow (\lambda-2) [(\lambda-4)^2 - 4] = 0$$

$$\Rightarrow (\lambda-2) [\lambda^2 - 2(\lambda)(4) + (4)^2 - 4] = 0$$

$$\Rightarrow (\lambda-2) [\lambda^2 - 8\lambda + 16 - 4] = 0$$

$$\Rightarrow (\lambda-2) [\lambda^2 - 8\lambda + 12] = 0$$

$$\lambda - 2 = 0$$

$$\boxed{\lambda = 2}$$

$$\lambda^2 - 8\lambda + 12 = 0$$

$$\lambda^2 - 6\lambda - 2\lambda + 12 = 0$$

$$\lambda(\lambda - 6) - 2(\lambda - 6) = 0$$

$$(\lambda - 6)(\lambda - 2) = 0$$

$$\lambda - 6 = 0$$

$$\lambda - 2 = 0$$

$$\boxed{\lambda = 6}$$

$$\boxed{\lambda = 2}$$

The Eigenvalue $\lambda = 2$ has algebraic Multiplicity = 2 & the eigenvalue $\lambda = 6$ has algebraic multiplicity = 1

For Eigen Space:-

Consider $(\lambda I - A)X = 0$

$$\begin{bmatrix} \lambda - 2 & 0 & 0 \\ -1 & \lambda - 4 & 1 \\ 2 & 4 & \lambda - 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 2$

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & -2 & 1 \\ 2 & 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now, convert it into Echelon Form

Take, $\begin{bmatrix} 0 & 0 & 0 \\ -1 & -2 & 1 \\ 2 & 4 & -2 \end{bmatrix}$

$$\begin{bmatrix} -1 & -2 & 1 \\ 2 & 4 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(-1)R_1$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_2 - 2R_1$$

So,
$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + 2x_2 - x_3 = 0 \rightarrow (i)$$

x_1 is leading variable

$$\boxed{x_2 = s}, \quad \boxed{x_3 = t}$$

Put in (i)

$$x_1 + 2s - t = 0$$

$$\boxed{x_1 = -2s + t}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2s + t \\ s \\ t \end{bmatrix}$$

$$= s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Here are two eigen vectors corresponding to eigen value $\boxed{\lambda = 2}$

The geometric multiplicity of eigenvalue is defined as dimension of eigen space associated with that eigen value.

Eigen vectors are

$$\boxed{\lambda_1 = 2} \quad \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

The dimension is 2.

The Geometric multiplicity of $\lambda = 2$ is 2.
For $\lambda = 6$

$$\begin{bmatrix} \lambda - 2 & 0 & 0 \\ -1 & \lambda - 4 & 1 \\ 2 & 4 & \lambda - 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 & 0 \\ -1 & 2 & 1 \\ 2 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Take, $\begin{bmatrix} -1 & 2 & 1 \\ 4 & 0 & 0 \\ 2 & 4 & 2 \end{bmatrix} \quad R_1 \approx R_2$

$$\begin{bmatrix} 1 & -2 & -1 \\ 4 & 0 & 0 \\ 2 & 4 & 2 \end{bmatrix} \quad (-1)R_1$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 8 & 4 \\ 0 & 8 & 4 \end{bmatrix} \quad \begin{array}{l} R_2 - 4R_1 \\ R_3 - 2R_1 \end{array}$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 8 & 4 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 - R_2$$

$$\begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix} \quad \frac{1}{8} R_2$$

$$x_1 - 2x_2 + x_3 = 0 \quad \longrightarrow \quad (i')$$

$$x_2 + \frac{1}{2}x_3 = 0 \quad \longrightarrow \quad (ii')$$

$$x_3 = t$$

$$x_2 = -\frac{1}{2}t$$

Put values in (ii),

$$x_1 - 2\left(-\frac{1}{2}t\right) - t = 0$$

$$x_1 + t - t = 0$$

$$\boxed{x_1 = 0}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{2}t \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

The dimension is 1.

The Geometric multiplicity of $\lambda = 6$ is 1.

(b) Determine whether A is diagonalizable?

Matrix A is diagonalizable if the geometric multiplicity of each eigen value is equal to algebraic multiplicity.

In our case, it is hold.

To find matrix P, write basis vectors in column form.

$$P_1 = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{bmatrix}, P_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, P_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$P = [P_1 \quad P_2 \quad P_3]$$

$$P = \begin{bmatrix} 0 & -2 & 1 \\ -\frac{1}{2} & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

(c) Prove that $P^{-1}AP = O$

Now, we find P^{-1}

$$\left[\begin{array}{ccc|ccc} 0 & -2 & 1 & 1 & 0 & 0 \\ -1/2 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 & 1 & 0 \\ 0 & -2 & 1 & 0 & 0 & 1 \end{array} \right] R_3 \approx R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1/2 & 0 & 1 & 1/2 \\ 0 & -2 & 1 & 1 & 0 & 0 \end{array} \right] R_2 + \frac{1}{2} R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1/2 & 0 & 1 & 1/2 \\ 0 & 0 & 2 & 1 & 2 & 1 \end{array} \right] R_3 + 2R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1/2 & 0 & 1 & 1/2 \\ 0 & 0 & 1 & 1/2 & 1 & 1/2 \end{array} \right] \frac{1}{2} R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1/4 & 1/2 & 1/4 \\ 0 & 0 & 1 & 1/2 & 1 & 1/2 \end{array} \right] R_2 - \frac{1}{2} R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/2 & -1 & 1/2 \\ 0 & 1 & 0 & -1/4 & 1/2 & 1/4 \\ 0 & 0 & 1 & 1/2 & 1 & 1/2 \end{array} \right] R_1 - R_3$$

$$\left[I \mid P^{-1} \right]$$

$$P^{-1} = \begin{bmatrix} -1/2 & -1 & 1/2 \\ -1/4 & 1/2 & 1/4 \\ 1/2 & 1 & 1/2 \end{bmatrix}$$

Now, $P^{-1}AP = D$

$$\begin{aligned} P^{-1}AP &= \begin{bmatrix} -1/2 & -1 & 1/2 \\ -1/4 & 1/2 & 1/4 \\ 1/2 & 1 & 1/2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{bmatrix} \begin{bmatrix} 0 & -2 & 1 \\ -1/2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -3 & -6 & 3 \\ -1/2 & 1 & 1/2 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 & -2 & 1 \\ -1/2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = D \end{aligned}$$

where diagonal entries are the eigen values of matrix A.

(d) Check that Matrix A & $P^{-1}AP$ have same trace.

Trace $(P^{-1}AP) =$ sum of diagonal entries

$$\begin{aligned} &= \text{trace} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \\ &= 6 + 2 + 2 \\ &= 10 \end{aligned}$$

$$\text{Trace}(A) = \text{tr} \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{bmatrix}$$

$$\begin{aligned} & 2 + 4 + 4 \\ & = 10 \end{aligned}$$

Both A & PAP have same trace.

b) Gram-Schmidt Process

$$u_1 = (1, 1, 1), u_2 = (-1, 1, 0), u_3 = (1, 2, 1)$$

$$v_1 = u_1 = (1, 1, 1) \Rightarrow \boxed{v_1 = (1, 1, 1)}$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$$

$$= (-1, 1, 0) - \frac{(-1+1+0)}{1^2+1^2+1^2} (1, 1, 1)$$

$$= (-1, 1, 0) - 0$$

$$\boxed{v_2 = (-1, 1, 0)}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$$

$$= (1, 2, 1) - \frac{(1+2+1)}{1+1+1} (1, 1, 1) - \frac{(-1+2+0)}{1+1+0} (-1, 1, 0)$$

$$= (1, 2, 1) - \frac{4}{3} (1, 1, 1) - \frac{1}{2} (-1, 1, 0)$$

$$\boxed{v_3 = \left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}\right)}$$

$$q_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{3}}(1, 1, 1) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$q_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{2}}(-1, 1, 0) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$$

$$q_3 = \frac{v_3}{\|v_3\|} = \frac{1}{\sqrt{6}}\left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}\right)$$

$$= \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)$$

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{bmatrix}$$

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \langle u_3, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \langle u_3, q_2 \rangle \\ 0 & 0 & \langle u_3, q_3 \rangle \end{bmatrix}$$

$$\begin{aligned}
 \langle u_1, q_1 \rangle &= (1, 1, 1) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \\
 &= (1) \left(\frac{1}{\sqrt{3}} \right) + (1) \frac{1}{\sqrt{3}} + (1) \left(\frac{1}{\sqrt{3}} \right) \\
 &= \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} = \frac{3}{\sqrt{3}}
 \end{aligned}$$

$$\begin{aligned}
 \langle u_2, q_1 \rangle &= (-1, 1, 0) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \\
 &= -\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} + 0 = 0
 \end{aligned}$$

$$\begin{aligned}
 \langle u_3, q_1 \rangle &= (1, 2, 1) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \\
 &= \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}}
 \end{aligned}$$

$$\begin{aligned}
 \langle u_2, q_2 \rangle &= (-1, 1, 0) \cdot \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \\
 &= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}}
 \end{aligned}$$

$$\begin{aligned}
 \langle u_3, q_2 \rangle &= (1, 2, 1) \cdot \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \\
 &= -\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} + 0 = \frac{1}{\sqrt{2}}
 \end{aligned}$$

$$\langle u_3, v_3 \rangle = (1, 2, 1) \cdot \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right)$$

$$= \frac{1}{\sqrt{6}} + \frac{2}{\sqrt{6}} - \frac{2}{\sqrt{6}}$$

$$= \frac{1}{\sqrt{6}}$$

so

$$R = \begin{pmatrix} \frac{3}{\sqrt{3}} & 0 & \frac{4}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{6}} \end{pmatrix}$$

Question No 5

- (a) Consider basis $S = \{v_1, v_2\}$ for $\mathbb{R}^2$, where $v_1 = (-2, 1)$ and $v_2 = (1, 3)$, and let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear transformation such that $T(v_1) = (-1, 2, 0)$ and $T(v_2) = (0, -3, 5)$. Find a formula for $T(x_1, x_2)$ and use formula to find $T(2, -3)$.

Sol.

Firstly, we have to find $T(x_1, x_2) = A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, where,

$$A = [T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \mid T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right)]$$

Now,

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = c_1 v_1 + c_2 v_2 = c_1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2c_1 + c_2 \\ c_1 + 3c_2 \end{pmatrix}$$

$$\Rightarrow \begin{aligned} -2c_1 + c_2 &= 1, & c_1 + 3c_2 &= 0 \\ -2c_1 - \frac{1}{3}c_1 &= 1 & c_1 &= -3c_2 \end{aligned}$$

$$\frac{-7c_1}{3} = 1$$

$$\Rightarrow c_1 = \frac{-3}{7}$$

$$\Rightarrow c_2 = -\frac{1}{3} \left(\frac{-3}{7} \right) = \frac{1}{7}$$

Thus,

$$\begin{aligned} T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) &= T(c_1 v_1 + c_2 v_2) = c_1 T(v_1) + c_2 T(v_2) \\ &= \frac{-3}{7} \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} + \frac{1}{7} \begin{pmatrix} 0 \\ -3 \\ 5 \end{pmatrix} \\ &= \left(\frac{3}{7}, \frac{-9}{7}, \frac{5}{7} \right) \end{aligned}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = a_1 v_1 + a_2 v_2 = a_1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} + a_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2a_1 + a_2 \\ a_1 + 3a_2 \end{pmatrix}$$

$$\Rightarrow -2a_1 + a_2 = 0, \quad a_1 + 3a_2 = 1$$

$$\Rightarrow a_2 = 2a_1, \quad a_1 + 3(2a_1) = 1$$

$$\Rightarrow a_2 = 2/7, \quad 7a_1 = 1$$

$$a_1 = 1/7$$

Thus,

$$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = T(a_1 v_1 + a_2 v_2) = a_1 T(v_1) + a_2 T(v_2)$$

$$= \frac{1}{7} \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} + \frac{2}{7} \begin{pmatrix} 0 \\ -3 \\ 5 \end{pmatrix}$$

$$= \left(-1/7, -4/7, 10/7 \right)$$

Now,

$$T(x_1, x_2) = \begin{bmatrix} \frac{3}{7} & -1/7 \\ -9/7 & -4/7 \\ 5/7 & 10/7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} (3x_1 - x_2)/7 \\ (-9x_1 - 4x_2)/7 \\ (5x_1 + 10x_2)/7 \end{bmatrix} \text{ Answer.}$$

(A)

For $T(2, -3)$ put $x_1 = 2, x_2 = -3$ in eqn (A).

$$T(2, -3) = \begin{bmatrix} (3(2) - (-3))/7 \\ (-9(2) - 4(-3))/7 \\ (5(2) + 10(-3))/7 \end{bmatrix} = \begin{bmatrix} 3/7 \\ 6/7 \\ -20/7 \end{bmatrix}$$

$$T(2, -3) = \left(3/7, 6/7, -20/7 \right) \text{ Answer.}$$

5(b)

let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation

$$\text{defined by } T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -4x_1 \\ -x_1 + 2x_2 \\ -2x_1 + 5x_2 \end{bmatrix}$$

(i)

Find the matrix for the transformation T i.e. $[T]_{B':B} = [[T(u_1)]_{B'} \mid [T(u_2)]_{B'}]$ relative to the basis $B = \{ \vec{u}_1, \vec{u}_2 \}$ and $B' = \{ \vec{v}_1, \vec{v}_2, \vec{v}_3 \}$ where

$$u_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, u_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, v_1 = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

$$v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Sol. $T(u_1) = c_1 v_1 + c_2 v_2 + c_3 v_3$

From the given formula

$$T(u_1) = T\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} -4(1) \\ -1+2(3) \\ -2(1)+5(3) \end{bmatrix} = \begin{bmatrix} -4 \\ 5 \\ 13 \end{bmatrix}$$

Now

$$(-4, 5, 13) = c_1(2, -2, 0) + c_2(1, 0, 1) + c_3(0, 1, 0)$$

$$\Rightarrow 2c_1 + c_2 = -4, \quad -2c_1 + c_3 = 5, \quad \boxed{c_2 = 13}$$

$$\Rightarrow c_1 = \frac{1}{2}(-4-13), \quad c_3 = 5 + 2\left(\frac{-17}{2}\right)$$

$$\boxed{c_1 = -17/2}$$

$$\boxed{c_3 = -12}$$

$$[T(u_1)]_{B'} = \begin{bmatrix} -17/2 \\ 13 \\ -12 \end{bmatrix}$$

Now, $T(u_2) = a_1 v_1 + a_2 v_2 + a_3 v_3$

From the given formula

$$T(u_2) = T\left(\begin{bmatrix} 2 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} -4(2) \\ -(2)+2(-1) \\ -2(2)+5(-1) \end{bmatrix} = \begin{bmatrix} -8 \\ -4 \\ -9 \end{bmatrix}$$

$$T(u_2) = (-8, -4, -9)$$

$$(-8, -4, -9) = a_1(2, -2, 0) + a_2(1, 0, 1) + a_3(0, 1, 0)$$

$$\Rightarrow 2a_1 + a_2 = -8, \quad -2a_1 + a_3 = -4, \quad a_2 = -9$$

$$a_1 = \frac{1}{2}(-8 + 9), \quad a_3 = -4 + 2\left(\frac{1}{2}\right)$$

$$\boxed{a_1 = \frac{1}{2}}, \quad \boxed{a_3 = 3}$$

$$\text{Thus, } [T(u_2)]_{B'} = \begin{bmatrix} \frac{1}{2} \\ -9 \\ 3 \end{bmatrix}$$

$$[T]_{B', B} = \begin{bmatrix} -\frac{17}{2} & \frac{1}{2} \\ 13 & -9 \\ -12 & 3 \end{bmatrix} = T$$

(ii) Find the Kernel of T i.e. $\text{Ker}(T)$.

$$\text{Ker}(T) = \{ (x_1, x_2) \mid T(x_1, x_2) = (0, 0, 0) \}$$

Consider,

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -\frac{17}{2} & \frac{1}{2} \\ 13 & -9 \\ -12 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} (-17x_1 + x_2)/2 \\ (13x_1 - 9x_2) \\ -12x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -17x_1 + x_2 = 0, \quad 13x_1 - 9x_2 = 0, \quad -12x_1 + 3x_2 = 0$$

$$x_2 = 17x_1, \quad 13x_1 - 9(17x_1) = 0$$

$$\Rightarrow \boxed{x_2 = 0}$$

$$-140x_1 = 0$$

$$\boxed{x_1 = 0}$$

$$\text{Ker}(T) = \{ (0, 0) \} \text{ Ans.}$$

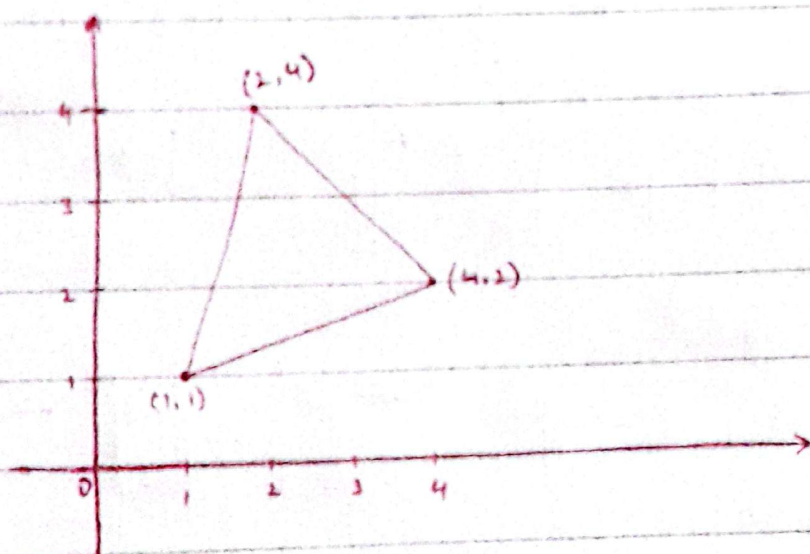
(iii) Since, by Rank-Nullity theorem:

$$\dim \text{Rang}(T) + \dim (\text{Ker}(T)) = \dim(\mathbb{R}^3)$$

$$\text{Rank}(T) + 0 = 3$$

$$\Rightarrow \text{Range}(T) = \mathbb{R}^3.$$

(c) If the following is the image of a triangle $\triangle MON$



Find the standard matrices for the mentioned below parts and plot the transformed image of a given triangle.

- i) Expand by a factor of 2 in the x-direction
let T be Transformation that is an expansion by a factor of 2 in the x-direction, then

$$\text{Transformation} \neq \text{in } \mathbb{R}^2$$

$$T(x, y) = (2x, y)$$

Now, for standard matrix A

$$T(1, 0) = (2, 0), \quad T(0, 1) = (0, 1)$$

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$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \text{ is the standard matrix.}$$

For plotting.

Given points of $\triangle MON$ are

$$(1, 1), (4, 2), (2, 4)$$

Now,

$$T(1, 1) = A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$T(1, 1) = (2, 1)$$

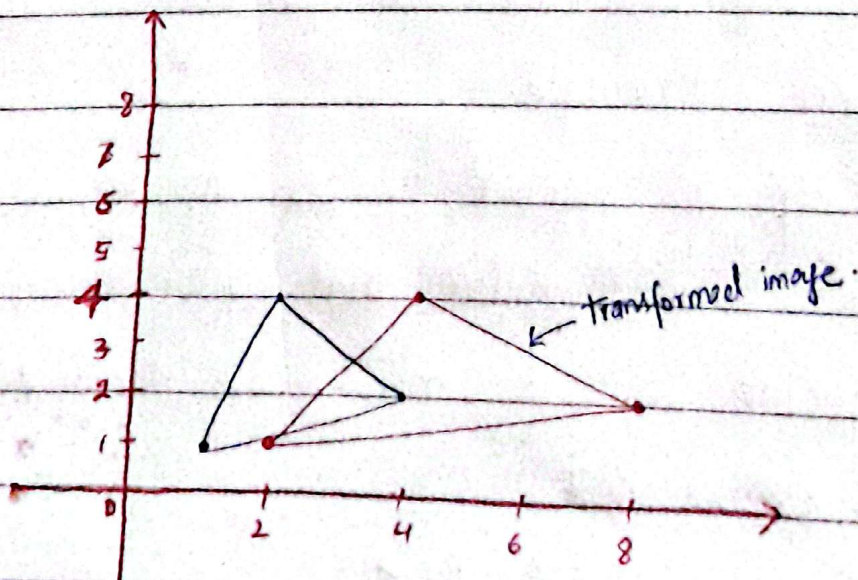
$$T(4, 2) = A \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

$$T(4, 2) = (8, 2)$$

$$T(2, 4) = A \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$T(2, 4) = (4, 4)$$

The transformed image of given triangle is



(ii) Reflect the given triangle about $y=x$.

If T is reflection about line $y=x$ then

$$T(x, y) = (y, x)$$

For Standard Matrix,

$$T(1, 0) = (0, 1), \quad T(0, 1) = (1, 0)$$

Thus, $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is the standard matrix

Now, Reflection of the given triangle $\triangle MON$.

$$T(1, 1) = A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$T(4, 2) = A \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$T(2, 4) = A \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Now, the transformed image is:

